

Indian Statistical Institute
M.Tech. (CS), First Year, Mid-Sem of Second Semester Examination, 2022-23
Database Management Systems

Full Marks: 30

Date: 03-03-2022

Time: 2 Hours

Answer any *three* of the following questions

$3 \times 10 = 30$

1. (a) Consider a pair of relations $R_1(X, Y)$ and $R_2(X, Y)$ having the number of tuples as $t_1 > 0$ and $t_2 > 0$, respectively. If the maximum and minimum number of tuples appearing in relations resulting from the operation $(R_1 - R_2) \cup (R_2 - R_1)$ differ by k , then show that the operation $(R_1 \cap R_2)$ will return at most $k/2$ tuples. Suppose X and Y take integer values only.
(b) Prove or disprove the statement: “For no pair of relations R_1 and R_2 more than one of the operations $R_1 \div R_2$, $R_1 - R_2$ and $R_1 \bowtie_{\theta} R_2$ are valid”.
(c) Cite an example where the selection and projection operations generate the same output from a particular relation.

5+3+2

2. (a) Suppose an AI startup wishes to create a database of the different modules and software prepared. A software may use one or more modules. The database must include the developers' details like employee ID as unique identifier, name and experience in months). Each software must have unique ID, developers' name, launch date, current version number and the modules they include. A software might be either free comprising license details or sellable having a price. Every module (may not be a part of any software or be a part of many software) must include a unique ID the language in which it is written, compatibility details, and developers' details. Construct an Entity Relationship diagram to represent an appropriate design for the said database.
(b) Convert the Entity Relationship diagram constructed in (a) into relations with appropriate primary keys mentioned therein.

6+4

3. (a) Show the steps of constructing a B-tree having order 4 that receives elements in the following order.

4, 14, 20, 2, 1, 3, 70, 5

Recall that a B-tree of order m can accommodate maximum $m - 1$ keys in the root and non-leaf nodes but m keys in leaf nodes.

- (b) Show the steps of re-structuring the B-tree constructed in (a) after deleting the key 20 followed by 2.
(c) Suggest the best possible data structure to keep tabular data in the physical level so that deletion of tuples within a given range of primary key values can be done efficiently. Justify your answer.

4+(2+2)+2

4. (a) Suppose the names of the actors and movies of Tamil cinema (aka Kollywood) are

stored in a database. The database comprises the following two tables with the schema defined therein. The keys corresponding to each table are underlined. Consider that the date format is DD-MM-YYYY.

ACTOR = <aid: integer, *name*: string, *age*: integer, *experience*: integer>
MOVIE = <mid: integer, aid: integer, *release*: date, *budget*: real>

Write the following queries in SQL.

- (i) Find the names of actors who have a middle name.
 - (ii) Find the names of actors who have an experience more than half of their age.
 - (iii) Find the movies that were released between 2022 and 2023.
 - (iv) Find the movies that are not yet released.
- (b) How can you change the primary key from *aid* to *name* in the table ACTOR using SQL?
(2+2+2+2)+2
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